

Discrete Mathematics and Graph Theory

Module 1 (CO1) — Logic and Relations

Previous Year Questions with Complete Solutions

C. V. Raman Global University, Odisha · B.Tech, 3rd Semester

Compiled from: MA203 (Regular 2024), MA108 (Regular 2025), MTH21105 (Back 2025), MA203 (Summer Quarter 2025)

This booklet collects every **CO1 (Module 1)** question from the four uploaded question papers and solves each one step by step. Part A questions carry 2 marks; Part B questions carry 7 marks.

Part A — Short Questions (2 Marks)

Q1. (2 marks) **Show that $P \vee \neg P$ is a tautology.**

Solution. A tautology is a compound proposition that is true for every assignment of truth values.

P	$P \vee \neg P$
T	T
F	T

The final column contains only T. Hence $P \vee \neg P$ is always true, i.e. it is a tautology. This is the *Law of Excluded Middle*.

Q2. (2 marks) **State De Morgan's law for quantifiers.**

Solution. The rules for negating quantified statements are:

$$\neg \forall x P(x) \equiv \exists x \neg P(x), \quad \neg \exists x P(x) \equiv \forall x \neg P(x).$$

In words: the negation of “for all” is “there exists a counterexample”, and the negation of “there exists” is “for all, not”. For example, the negation of “Every student passed” is “There exists a student who did not pass”.

Q3. (2 marks) **Find the maximal and minimal elements of the partially ordered set $(\{1, 2, 3, 4, 6, 8\}, |)$.**

Solution. The relation is divisibility. An element is *minimal* if no other element strictly divides it, and *maximal* if no other element in the set is a proper multiple of it.

- **Minimal:** 1 divides every element and nothing divides 1, so 1 is minimal. Every other element (2, 3, 4, 6, 8) has 1 below it, so none of them is minimal. $\Rightarrow$ **minimal element** = 1.
- **Maximal:** 6 has no proper multiple in the set (12, 18, ... are absent) and 8 has none (16, 24 are absent). But $2 \mid 4$, $4 \mid 8$, $3 \mid 6$ etc., so 2, 3, 4 are not maximal. $\Rightarrow$ **maximal elements** = 6 and 8.

Q4. (2 marks) **Construct the truth table for the relation $(P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)$.**

Solution.

P	Q	$P \rightarrow Q$	$\neg Q$	$\neg P$	$\neg Q \rightarrow \neg P$	$(P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)$
T	T	T	F	F	T	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

The last column is T in every row, so the given proposition is a **tautology**. This simply restates the fact that a conditional statement is logically equivalent to its contrapositive: $(P \rightarrow Q) \equiv (\neg Q \rightarrow \neg P)$.

Q5. (2 marks) Use mathematical induction to prove that $n^2 + n$ is divisible by 2 for all $n \in \mathbb{N}$.

Solution. Let $P(n)$: “ $n^2 + n$ is divisible by 2”.

- **Basis** ($n = 1$): $1^2 + 1 = 2$, which is divisible by 2. So $P(1)$ is true.
- **Induction hypothesis:** assume $P(k)$ is true, i.e. $k^2 + k = 2m$ for some integer m .
- **Inductive step:**

$$(k+1)^2 + (k+1) = k^2 + 2k + 1 + k + 1 = (k^2 + k) + 2(k+1) = 2m + 2(k+1) = 2(m+k+1),$$

which is divisible by 2. So $P(k+1)$ is true.

By the principle of mathematical induction, $n^2 + n$ is divisible by 2 for all $n \in \mathbb{N}$. ■

Q6. (2 marks) Find the transitive closure of the relation $R = \{(1,2), (2,3)\}$ on the set $\{1,2,3\}$.

Solution. The transitive closure is $t(R) = R \cup R^2 \cup R^3 \cup \dots$.

$$R = \{(1,2), (2,3)\}, \quad R^2 = R \circ R = \{(1,3)\}, \quad R^3 = R^2 \circ R = \emptyset.$$

Hence

$$t(R) = R \cup R^2 = \{(1,2), (2,3), (1,3)\}.$$

(In matrix form, $M_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ and its transitive closure is $\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$.)

Q7. (2 marks) Show that $q \rightarrow (\neg p \vee q)$ is a tautology.

Solution. Simplify using $A \rightarrow B \equiv \neg A \vee B$:

$$q \rightarrow (\neg p \vee q) \equiv \neg q \vee (\neg p \vee q) \equiv (\neg q \vee q) \vee \neg p \equiv T \vee \neg p \equiv T.$$

Since it reduces to T , it is a tautology. The truth table confirms this:

p	q	$\neg p \vee q$	$q \rightarrow (\neg p \vee q)$
T	T	T	T
T	F	F	T
F	T	T	T
F	F	T	T

Q8. (2 marks) Find the maximal and minimal elements of the partially ordered set $(\{2, 4, 5, 10, 12, 20, 25\}, |)$.

Solution.

- **Minimal:** 2 and 5 have no other element of the set dividing them (the set has no 1), so they are minimal. But $2 \mid 4$, $2 \mid 10$, $5 \mid 10$, $5 \mid 25$, $2 \mid 12$, $2 \mid 20$, so 4, 10, 12, 20, 25 are not minimal. $\Rightarrow$ **minimal elements** = 2, 5.
- **Maximal:** 12, 20 and 25 have no proper multiple within the set, so they are maximal. Since $2 \mid 4$, $4 \mid 12$, $10 \mid 20$, etc., the rest are not maximal. $\Rightarrow$ **maximal elements** = 12, 20, 25.

Q9. (2 marks) Let n be an integer and $3n + 2$ be odd. Show that n is odd by the method of contrapositive.

Solution. The contrapositive of “ $3n + 2$ odd $\Rightarrow n$ odd” is “ n even $\Rightarrow 3n + 2$ even”. We prove this. Assume n is even, so $n = 2k$ for some integer k . Then

$$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1),$$

which is even. Hence, whenever n is even, $3n + 2$ is even. By the contrapositive equivalence, if $3n + 2$ is odd then n must be odd. ■

Part B — Long Questions (7 Marks)

Q10. (7 marks) Prove using a truth table that $(P \vee Q) \wedge (\neg P \vee R)$ is logically equivalent to $(P \wedge R) \vee (Q \wedge \neg R)$.

Solution.

P	Q	R	$P \vee Q$	$\neg P \vee R$	$(P \vee Q) \wedge (\neg P \vee R)$	$(P \wedge R) \vee (Q \wedge \neg R)$
T	T	T	T	T	T	T
T	T	F	T	F	F	T
T	F	T	T	T	T	T
T	F	F	T	F	F	F
F	T	T	T	T	T	F
F	T	F	T	T	T	T
F	F	T	F	T	F	F
F	F	F	F	T	F	F

Comparing the last two columns, they agree in 5 of the 8 rows but **differ** in rows (T, T, F) , (F, T, T) and (F, T, F) . So, exactly as printed, the two propositions are **not** logically equivalent.

Note. The right-hand side appears to contain a typo; the correct standard equivalence is

$$(P \vee Q) \wedge (\neg P \vee R) \equiv (P \wedge R) \vee (\neg P \wedge Q),$$

which is verified below (the last two columns now match in every row):

P	Q	R	$P \vee Q$	$\neg P \vee R$	$(P \vee Q) \wedge (\neg P \vee R)$	$(P \wedge R) \vee (\neg P \wedge Q)$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	T	T	T	T
T	F	F	T	F	F	F
F	T	T	T	T	T	T
F	T	F	T	T	T	T
F	F	T	F	T	F	F
F	F	F	F	T	F	F

In the exam, if the printed form is asked, show the truth table and state the correct equivalent form as above.

Q11. (7 marks) **Apply Warshall's algorithm to compute the transitive closure of the relation represented by the adjacency matrix**

$$M = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}.$$

Solution. Warshall's rule: $W^k[i][j] = 1$ if $W^{k-1}[i][j] = 1$, or if both $W^{k-1}[i][k] = 1$ and $W^{k-1}[k][j] = 1$.

$$W^0 = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \quad W^1 = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \quad W^2 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

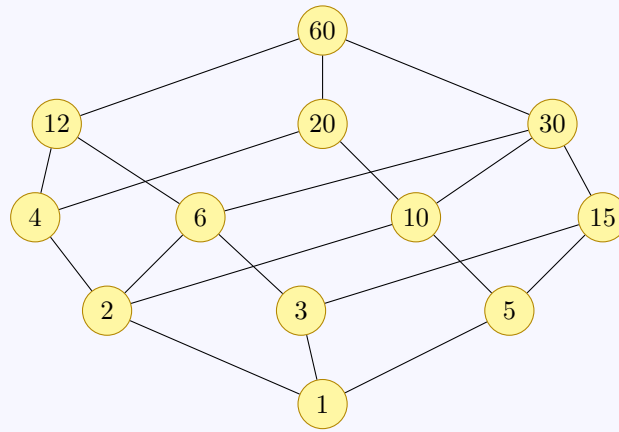
$$W^3 = W^4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

Since $W^4 = W^3$, the algorithm terminates. The transitive closure is the **universal relation**: every element is related to every element.

Q12. (7 marks) **Define a poset. Draw the Hasse diagram for the divisors of 60 under the “divides” relation and verify whether that forms a poset.**

Solution. Definition. A *poset* (partially ordered set) is a set S together with a relation $\preceq$ that is reflexive, antisymmetric and transitive, written $(S, \preceq)$.

The divisors of 60 are 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60. Their Hasse diagram (loops and transitive edges removed, edges drawn upwards) is:



Verification that $(\{1, 2, \dots, 60\}, |)$ is a poset:

- **Reflexive:** $a \mid a$ for every divisor a .
- **Antisymmetric:** if $a \mid b$ and $b \mid a$ then $a = b$.
- **Transitive:** if $a \mid b$ and $b \mid c$ then $a \mid c$.

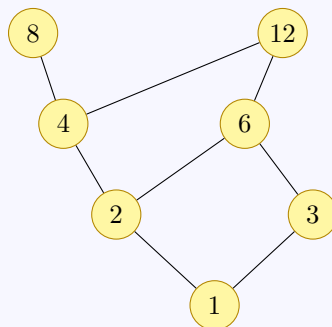
All three hold, so the divisibility relation is a partial ordering and $(D(60), |)$ is a poset. (It is in fact a lattice, since every pair has a gcd and an lcm.)

Q13. (7 marks) **Show that the division relation on the set of integers is a partial ordering relation. Draw the Hasse diagram representing the partial ordering $\{(a, b) : a \text{ divides } b\}$ on $\{1, 2, 3, 4, 6, 8, 12\}$.**

Solution. On the set of positive integers the divisibility relation $|$ is:

- **Reflexive:** $a \mid a$ for every integer a .
- **Antisymmetric:** if $a \mid b$ and $b \mid a$ then $a = b$.
- **Transitive:** if $a \mid b$ and $b \mid c$, then $b = ak$ and $c = bl$ for integers k, l , so $c = a(kl)$, i.e. $a \mid c$.

Hence $(\mathbb{Z}^+, |)$ is a poset. *Caveat:* over *all* integers (including negatives) antisymmetry fails, e.g. $2 \mid -2$ and $-2 \mid 2$ but $2 \neq -2$; so the division relation is a partial ordering on the positive integers. The Hasse diagram for $\{1, 2, 3, 4, 6, 8, 12\}$:



Here 1 is the least element, and 8 and 12 are the maximal elements.

Q14. (7 marks) **Find the transitive closure of the relation $R = \{(1, 2), (2, 1), (2, 3), (3, 4)\}$ on $A = \{1, 2, 3, 4\}$ using Warshall's algorithm.**

Solution. The initial matrix is

$$W^0 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Applying $W^k[i][j] = 1$ if $W^{k-1}[i][j] = 1$ or ($W^{k-1}[i][k] = 1$ and $W^{k-1}[k][j] = 1$):

$$W^1 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad W^2 = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad W^3 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad W^4 = W^3.$$

Hence

$$t(R) = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4), (3, 4)\}.$$

Q15. (7 marks) **Prove that $n^3 - n$ is divisible by 3 for every positive integer n by using mathematical induction.**

Solution. Let $P(n)$: “ $n^3 - n$ is divisible by 3”.

- **Basis** ($n = 1$): $1^3 - 1 = 0 = 3 \times 0$, divisible by 3. So $P(1)$ is true.
- **Induction hypothesis:** assume $P(k)$ is true, i.e. $k^3 - k = 3m$ for some integer m .
- **Inductive step:**

$$(k+1)^3 - (k+1) = (k^3 + 3k^2 + 3k + 1) - (k+1) = (k^3 - k) + 3(k^2 + k) = 3m + 3(k^2 + k) = 3(m + k^2 + k).$$

Both terms are multiples of 3, so the sum is divisible by 3. Thus $P(k + 1)$ is true.

By the principle of mathematical induction, $n^3 - n$ is divisible by 3 for every positive integer n . ■

Note on coverage. In the uploaded scan, page 2 of the first paper (MA203, Regular 2024) — carrying its Part B questions Q2–Q4 — was not captured; only its Part A (Q1a–Q1c) is available. All Module 1 questions from the remaining papers are included above.

Quick pointers for Module 1: the most repeated long questions are *Warshall’s algorithm / transitive closure* (Q11, Q14) and the *POSET–Hasse diagram* question (Q12, Q13); among short questions, *truth tables*, *induction*, and *maximal/minimal elements* recur most often.